

COLE CAMPBELL

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SUMMARY

Recently graduated *cum laude* as a Data Scientist and Machine Learning Engineer, with experience in statistical modeling, deep learning, and AI applications. Material has been learned through the classroom and applied through projects: Three projects engaging in the cleaning, wrangling, and modeling of data; one project on the topic of deep learning; two projects concentrated on AI in applications. Pursuing an M.S. in Data Science on top of a B.S. in Data Science, completed May 2026.

EDUCATION

Arizona State University: Tempe, AZ

M.S. in Data Science, Analytics & Engineering (Computational Mathematics & Data); 08/2026 to 05/2027

B.S. in Data Science (Computer Science Concentration); 08/2023 to 05/2026, GPA 3.53 / 4.00

Barrett, The Honors College; 08/2023 to 05/2026

TECHNICAL SKILLS

Languages: Python, R, Dart, C/C++, JavaScript (Node.js), SQL, Bash, MATLAB

ML / AI: Deep learning (CNNs, RNNs/LSTMs, Transformers), generative models (Diffusion, GANs, StyleGAN), LLMs & prompt engineering, LoRA/QLoRA/PEFT, transfer learning, neural style transfer, audio/music generation, model evaluation (FID, CLIP, LPIPS, BLEU, perplexity)

Frameworks: PyTorch (torchvision/torchaudio), TensorFlow, Hugging Face (Transformers, Diffusers, Accelerate), scikit-learn, NumPy, Pandas, SciPy, StatsModels, FastAPI, Flutter/Riverpod

Data Science: Regression & statistical modeling, classification, time series, Bayesian, clustering, longitudinal/panel analysis, feature engineering, A/B testing, causal inference, hypothesis testing, visualization (Matplotlib, Seaborn, Plotly)

Data Engineering: ETL & data pipelines, REST API integration, SQLAlchemy + Alembic, PostgreSQL, Firebase, Docker, CUDA, cloud (Google Cloud Console)

Tools: Git, Linux, Jupyter, VS Code, OpenCV, librosa, ParaView/VTK, pytest, Weights & Biases, Kaggle/NBA APIs, REST/OpenAPI

Math: Linear algebra, probability, optimization, numerical methods, stochastic processes, statistical theory

PROJECTS

Tempo; AI Scheduling Assistant (2026)

FastAPI • Flutter • PostgreSQL • LLM / NLP • Constraint Optimization

- Built a cross-platform AI scheduling app (Flutter/FastAPI) with a constraint-based engine that generates, compares, and applies alternative day/week schedule plans.
- Added a natural-language “Ask Tempo” interface and a scenario pipeline supporting diff-based comparison and tradeoff analysis.
- Backed by SQLAlchemy/Alembic and Firebase auth; supports continuous testing with 360+ backend tests at about 84% coverage.

Multi-Architecture Generative Content Studio; Deep Learning Capstone (2026)

PyTorch • Transformers • Diffusion • GANs • CNNs • RNNs

- Built a unified generative pipeline producing story text, illustrations, character portraits, and music from a single prompt, integrating five architectures (GPT-2, Stable Diffusion, StyleGAN2/3, LSTM) into one system.
- Fine-tuned Stable Diffusion (LoRA + ControlNet) and StyleGAN2 (GAN inversion, latent editing) for controlled scenes and character consistency.
- Benchmarked DCGAN against Stable Diffusion using FID/CLIP metrics; Stable Diffusion had 390x better image quality.

NBA Shot Quality Capstone; Applied Machine Learning & Sports Analytics Project (2026)

Python • Machine Learning • Statistical Analysis

- Led a four-person team to build an end-to-end Machine Learning pipeline analyzing 2.9 million NBA shot attempts (2013-2025) to separate shot-making skill from shot selection; found that shot-quality-adjusted player rankings diverge from traditional metrics (Spearman's $\rho = 0.39$ vs. FG%), with top performers shifting up to 195 ranking spots once shot difficulty is accounted for.

EXPERIENCE

Handshake AI: Remote

AI Trainer / Technical Evaluator (Contract); 04/2026 to Present

- Evaluate and refine AI model outputs across coding, STEM, and Data Science domains; stump the model, assess the failure, and provide feedback.

Spinato's Pizzeria: Tempe, AZ

Chef / Runner / Driver, 01/2024 to 04/2026

- Became cross-trained through every kitchen role within the first six months, showcasing consistency, accuracy, and eagerness to learn.